



**CATHOLIC JUNIOR COLLEGE**  
**JC2 PRELIMINARY EXAMINATION**  
**Higher 1**

**ANSWER SCHEME**

CANDIDATE  
NAME

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CLASS

2T

INDEX  
NUMBER

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**BIOLOGY**

**Paper 2 Structured and Free-response Questions**

**8876/02**

**24 AUGUST 2021**

**2 hours**

Candidates answer on Question Paper.  
No Additional Material are required.

**READ THESE INSTRUCTIONS FIRST**

Write your name, class and index number in the spaces at the top of this page.  
Write in dark blue or black pen.  
You may use an HB pencil for any diagrams or graphs.  
Do not use staples, paper clips, glue or correction fluid.

**Section A**

Answer **all** questions in the spaces provided on the Question Paper.

**Section B**

Answer any **one** question in the spaces provided on the Question Paper.

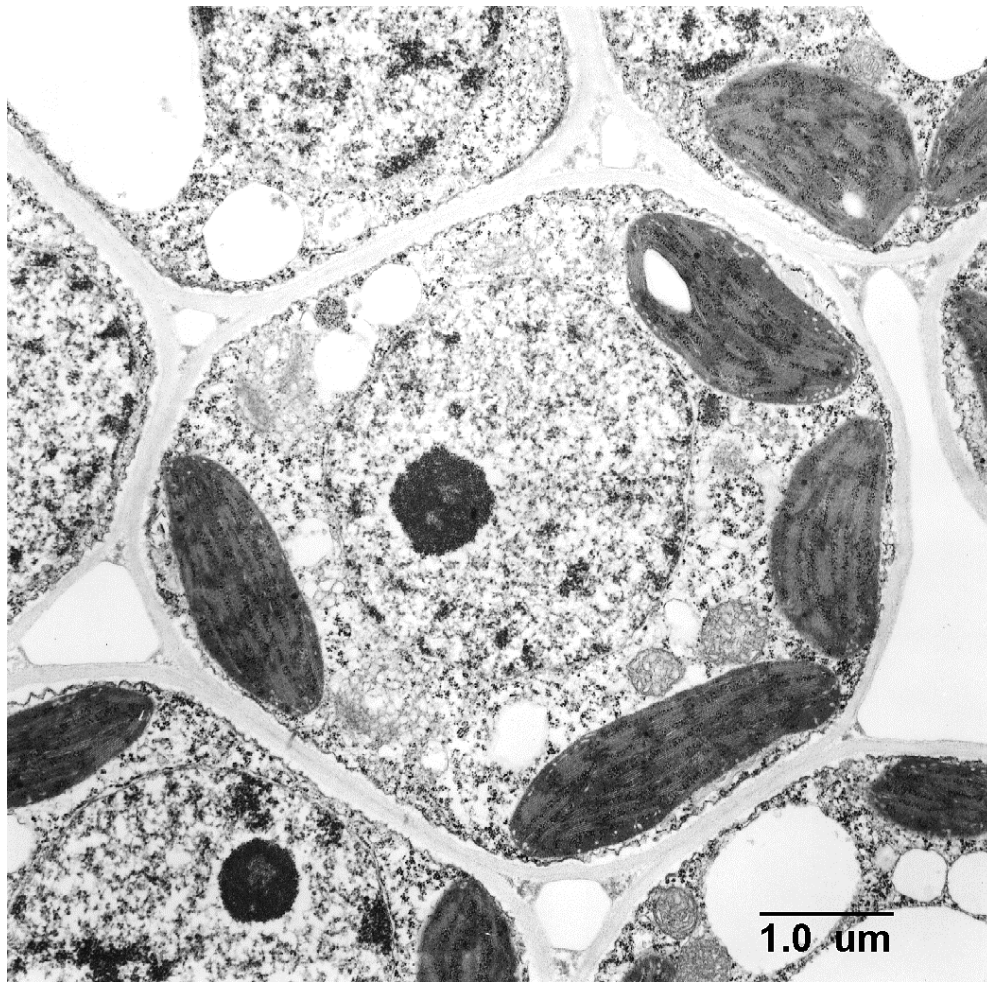
The use of an approved scientific calculator is expected, where appropriate.  
You may lose marks if you do not show your working or if you do not use appropriate units.

The number of marks is given in brackets [ ] at the end of each question or part question.

| For Examiner's Use |    |
|--------------------|----|
| Section A          |    |
| 1 [12]             |    |
| 2 [5]              |    |
| 3 [11]             |    |
| 4 [10]             |    |
| 5 [7]              |    |
| Section B          |    |
| 6 [15]             |    |
| 7 [15]             |    |
| TOTAL              | 60 |

Answer **all** questions.

- 1 Fig. 1.1 shows an electron micrograph of a cell.



**Fig. 1.1**

- (a) With reference to the Fig 1.1, state two reasons why the cell is considered to be a eukaryotic cell. Use 2 separate label lines, labelled **1** and **2**, to clearly mark out features visible in Fig. 1.1 to support your answer.

1. ....
  2. ....
- [2]

1. Presence of a **true, membrane-bound nucleus** (that encloses the genetic material), unlike a prokaryotic cell where the genetic material lies freely in the cytoplasm (in a denser region called the nucleoid)
2. Presence of **membrane-bound organelles** whereas these organelles are absent in a prokaryotic cell

Labels to point to the following:

1. Nucleus
2. Chloroplast or mitochondrion

*[Reject: if label numbering and answer does not match.]*

Stachyose is a storage oligosaccharide found in beans, peas and other legumes.

Stachyose can only be metabolized by anaerobic microorganisms in the large intestine. It is responsible for the production of carbon dioxide, hydrogen sulfide and other gases which collectively make up what is known as flatulence.

The structure of stachyose is shown in Fig. 1.2.

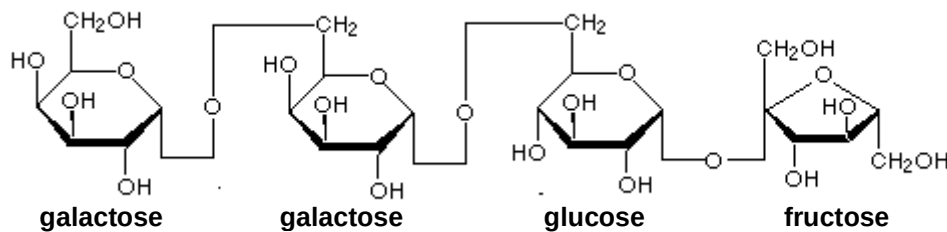


Fig. 1.2

(b) Define the term *glycosidic bond*.

.....  
 ..... [1]

1. A glycosidic bond is formed between **2 sugar residues / molecules** via a **condensation reaction** / removal of water;
2. when the anomeric **hydroxyl group of one sugar** unit react with another / any **hydroxyl group on another** sugar molecule

(c) Several commercial products are available to assist in the digestion of oligosaccharides like stachyose. Suggest how such products prevent the production of flatulence.

.....  
 ..... [1]

1. These products allow **the hydrolysis / cleavage / breakdown** of glycosidic bonds present in the oligosaccharides via **addition of a water molecule** before they become available to intestinal bacteria / anaerobic microorganisms;

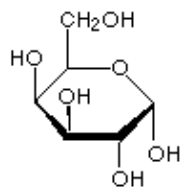
(d) When small amounts of paper containing cellulose are ingested by children, there is usually no gas production. Suggest one reason and a possible consequence of such an event.

.....  
 .....  
 ..... [2]

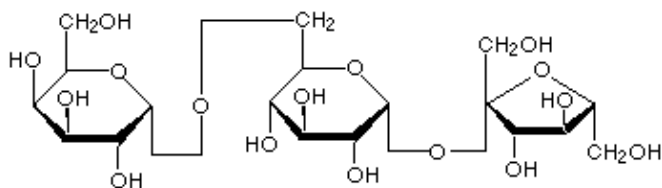
1. Cellulose is made up of  **$\beta$ -glucose** residues joined by  **$\beta$ -1,4 glycosidic bonds** which is not / cannot be broken down / digested in human intestines as humans **lack the enzyme cellulase**;
2. Intestinal blockage could occur OR The paper could pass out undigested or AVP

An enzyme known as  $\alpha$ -galactosidase cleaves  $\alpha$ -(1,6) glycosidic bonds to sequentially release terminal galactose residues. The breakdown of stachyose by  $\alpha$ -galactosidase is a two-step process. In the first step of this process, stachyose is hydrolysed to galactose and raffinose.

(e) With reference to Fig. 1.2, draw the products of the **first step in this hydrolysis**.



**Galactose**

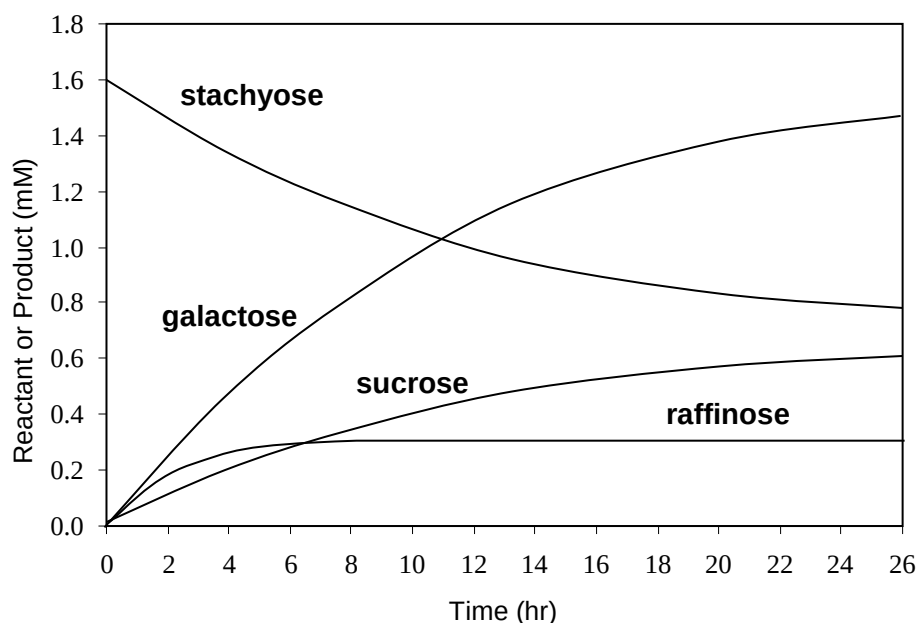


**Raffinose**

1. Galactose correctly drawn and labeled
2. Raffinose correctly drawn and labeled

[2]

Fig. 1.3 illustrates the activity of  $\alpha$ -galactosidase on the substrate stachyose over a period of time.



**Fig. 1.3**

(f) With reference to Fig. 1.3, explain why the level of raffinose remains relatively constant.

.....

.....

.....

..... [2]

1. Stachyose is broken down into galactose and raffinose as shown by the drop in stachyose concentration from 1.6 mM to ~0.8 mM from 0 to 26 hours;
2. In the second step of hydrolysis,  $\alpha$ -galactosidase is also able to hydrolyse / break down the  $\alpha$ -1,6 glycosidic bond in raffinose, producing both galactose and sucrose as evidenced by the concentration of sucrose increasing from 0mM to ~0.6 mM from 0 to 26 hours;
3. At net equilibrium, both reactions produces a relatively constant raffinose concentration of 0.3 mM;

- (g) Suggest why the rate at which stachyose is broken down decreases as the concentration of galactose increases.  
cause

.....  
 .....  
 .....  
 ..... [2]

1. Galactose acts as a competitive inhibitor which is structurally similar to the stachyose substrate molecule and/or complementary in shape to the active site of the enzyme
2. It competes with stachyose for access / by binding to the active site; hence slowing down the rate of stachyose hydrolysis.

OR

3. Galactose acts as a non-competitive inhibitor / end product inhibitor where it binds to an allosteric site  
and changes the three dimensional conformation of the active site of some  $\alpha$ -galactosidase enzyme molecules
4. Such that there is a reduction of effective  $\alpha$ -galactosidase /  $\alpha$ -galactosidase can no longer bind to stachyose; hence slowing down the rate of stachyose hydrolysis.

[Total: 12]

- 2 (a) The “trombone” model of DNA replication postulates that two DNA polymerase enzymes work together in a protein complex.

In Fig. 2.1, structures labelled **P**, **Q** and **R** are enzymes involved in the process of DNA replication. Connecting proteins join the two DNA polymerase molecules to enzyme **P**, forming the protein complex.

Key:

- parental DNA strand
- daughter DNA strand
- - - RNA

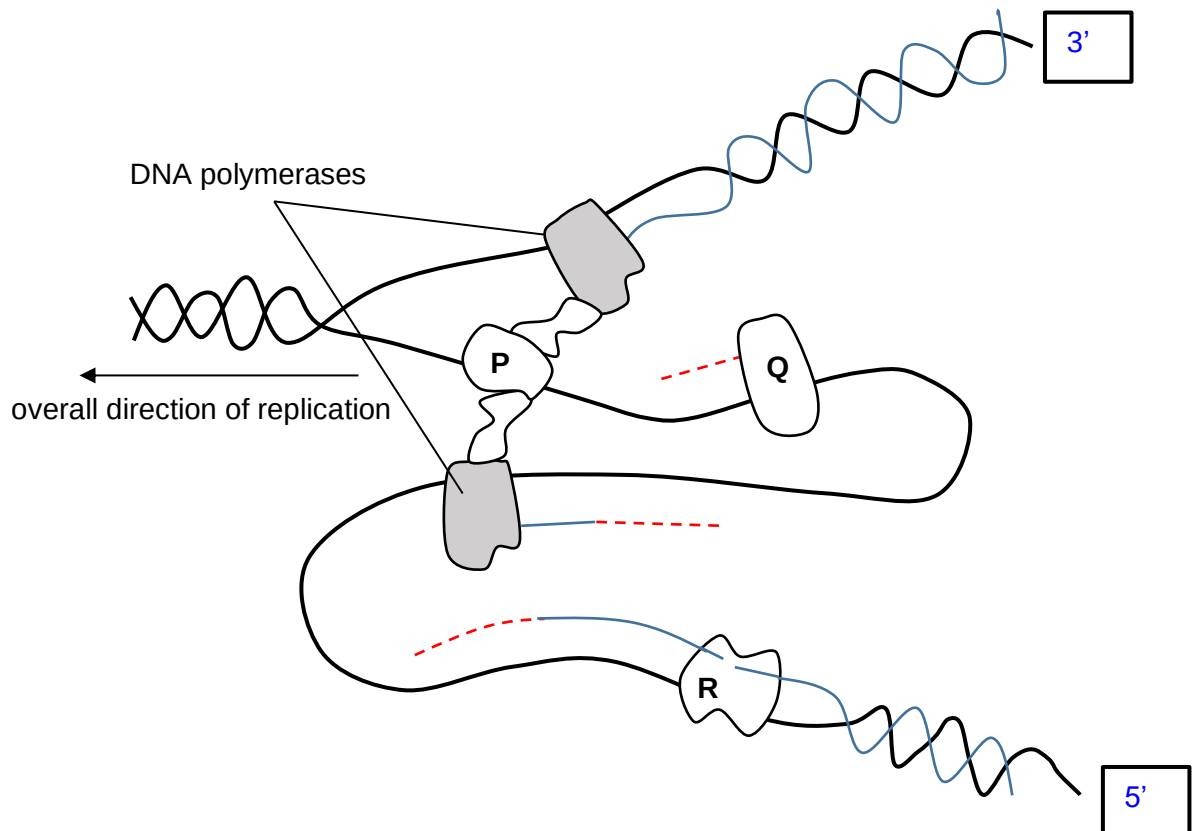


Fig. 2.1

- (i) Complete the boxes to indicate clearly the 5' and 3' ends of the parental strands shown in Fig. 2.1. [1]
- (ii) Identify enzymes **P**, **Q** and **R**. [2]

**P:** .....

**Q:** .....

**R:** .....

**P:** DNA Helicase

**Q:** Primase

**R:** DNA Ligase

Mark Allocation:

- 2 marks for all 3 correct answers
- 1 mark for any two correct answers

Fig 2.2 illustrates the binding of a protein to a certain region of a DNA molecule

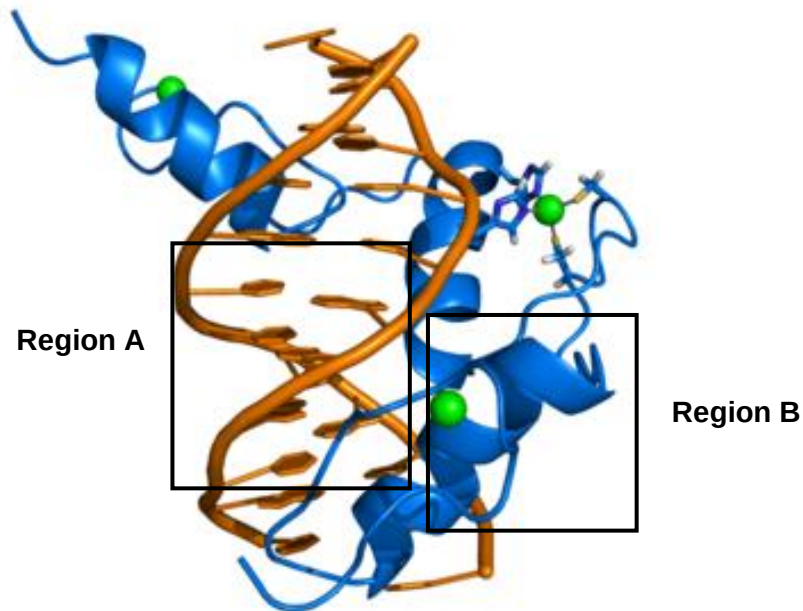


Fig 2.2

- (b) With reference to Fig. 2.2, describe how the helical structure shown in **Region A** differs from that of **Region B**.

.....

.....

.....

.....

.....

..... [2]

1. Helical structure (of the DNA) in region A is **double-stranded** unlike the **single-stranded** (polypeptide chain) in region B.
2. For Region A, each turn of the helix is made up of **10 base pair**; for Region B, the helix has **3.6 amino acids per turn**.
3. Helical structure in Region A is maintained by **phosphodiester linkages between deoxyribonucleotides** and **hydrogen bonds between complementary bases** whereas the helical structure of Region B by **peptide linkages** and **hydrogen bonds between amino acids**

[Total: 5]



- 3 Fig. 3.1 shows the structure of molecules and some associated processes found in a eukaryotic cell.

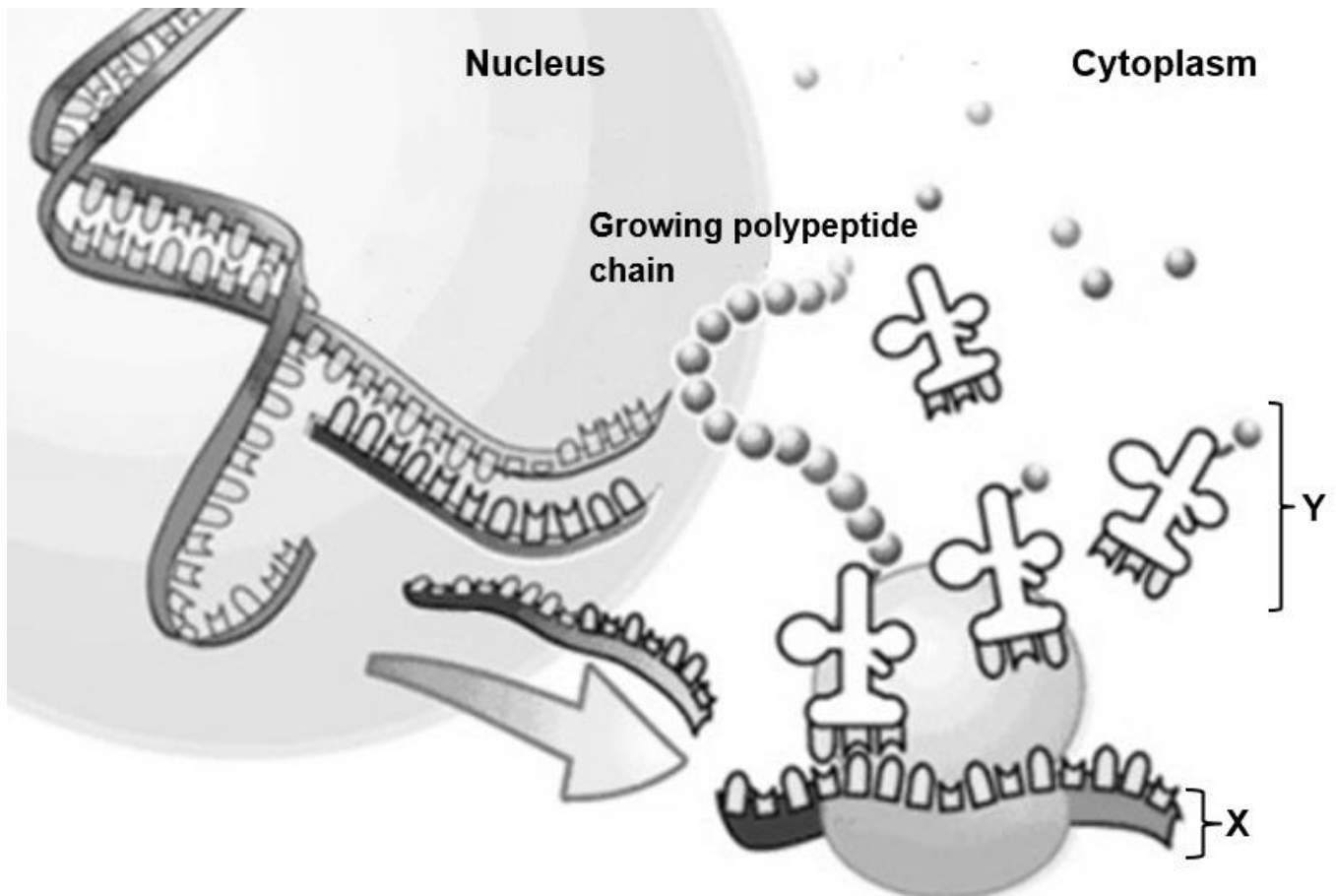


Fig. 3.1

- (a) With reference to Fig. 3.1, outline the events that has occurred in the nucleus that synthesised structure X.

.....

.....

.....

.....

.....

.....

.....

..... [4]

1. **RNA polymerase II** binds to **promoter** of specific gene, forming a **transcription initiation complex** together with **general transcription factors**;
2. (a) RNA polymerase **reads the DNA template strand from 3' to 5' direction** to form **pre-mRNA** / adds **ribonucleotides** to the **3'OH end** of the **pre-mRNA strand** or in **5' to 3' direction**;

OR



(b) RNA nucleotides / ribonucleotides are added via **complementary base pairing** with the **DNA template**;

OR

(c) **Phosphodiester bonds** are formed between ribonucleotides to form pre-mRNA;

3. (a) RNA **polymerase** continues with the transcription process until the **transcription stop site** is reached where the enzyme subsequently **dissociates** from **RNA-DNA**.

OR

(b) RNA polymerase II transcribes a **3'-TTATTT-5' sequence** (= the **polyadenylation signal**) on the DNA into **5'-AAUAAA-3'** along the **pre-mRNA**. At a point about **10 to 35 nucleotides downstream from the AAUAAA signal**, proteins that are associated with the growing RNA transcript cut the RNA transcript free from the RNA polymerase. This releases the **pre-mRNA**.

4. **RNA processing + (any 1 of the 3): 7-methylguanose / modified guanine cap** added to the **5' end** of the pre-mRNA strand / **3' poly(A) tail** added to the **3' end of the pre-mRNA** / **RNA splicing** to synthesise **mRNA**.

(b) Identify structure Y.

..... [1]

1. **Aminoacyl-tRNA / charged tRNA**

- (c) Parasites cause diverse types of disease, for example, malaria. Drug treatments are required to address varying causes of pathogenesis. A new class of drugs has been deployed to block the formation of structure Y in parasites.

Suggest how the use of such drugs is effective in treating parasites infections.

.....

..... [1]

1. Specific amino acids cannot be brought to the ribosomes, protein synthesis cannot occur causing death of parasites.

Epidermal growth factor (EGF) is a protein composed of 53 amino acids. It binds to specific cell surface membrane receptors to promote cell proliferation. An example of these surface membrane receptors is the HERS receptor which is a glycoprotein found on the cell surface membrane.

- (d) (i) Based on the information above, explain why EGF cannot move across cell surface membrane through channel proteins.

.....  
 .....  
 ..... [1]

1. EGF is composed of 53 amino acids which is too large in size to pass through the pore/hydrophilic channel of any channel proteins.

- (ii) Explain how the structure of HER2 receptor allows it to be embedded across cell surface membrane and serve as a transmembrane protein.

.....  
 .....  
 .....  
 .....  
 .....  
 .....  
 ..... [4]

1. The transmembrane region of HER2 receptor composed of amino acids with non-polar / hydrophobic R-group;
2. allowing the transmembrane region of HER2 receptor to interact via hydrophobic interaction with the fatty acids / hydrocarbon tail of phospholipid molecules.
3. The regions of HER2 receptor that are facing the aqueous intracellular and extracellular environment composed of amino acids with either polar or charged / hydrophilic R-group;
4. allowing these regions of HER2 receptor to interact via hydrophilic interaction with the environment / with the phosphate heads of the phospholipid molecules.

[Total: 11]

4 About one third of the leg injuries to racehorses involve tendon damage.

In 2006, bone marrow stem cells were taken from injured racehorses and cultured so that they divided many times by mitosis. Each horse's cells (usually around 10 million cells for one tendon) were then injected into its damaged tendons.

80% of the treated horses returned to racing, compared with 30% of those treated conventionally.

- (a) (i) Stem cells provide the means of maintenance and repair of tissue. With reference to the mitotic cell cycle, explain how the initiation of cell division is controlled.

.....  
 .....  
 .....  
 ..... [2]

1. Initiation of cell division is controlled by the **G1 Checkpoint**
2. **Cell division is controlled** by the presence of the **growth factors** such as growth hormone
3. And is inhibited by contact inhibition. (less significant)

Max 2

- (ii) Explain how it is possible that bone marrow stem cells could differentiate into the range of cell types needed for repairing the tendon injuries.

.....  
 .....  
 .....  
 ..... [2]

- 1 ref. bone marrow stem cells as **multipotent**; / able to differentiate into cells of **same/specific lineage**;
- 2 ref. bone marrow stem cells able to **respond to specific molecular signals** to differentiate;
- 3 AVP: stem cells containing a complete set of genome;

- (iii) Suggest an advantage of the above stem cell therapy.

.....  
 ..... [1]

- 1 Eliminates the risk of tissue/donor rejection if the cells were taken from another donor.
- 2 ref. stem cells **capable of continual self-renewal** and can be used to **expand cell numbers** in vitro / in the lab;

- (iv) Similar to stem cells, cancer cells are also capable to dividing many times via mitosis.  
State 2 differences between stem cells and cancer cells.

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..... [2]

Any two of the following:

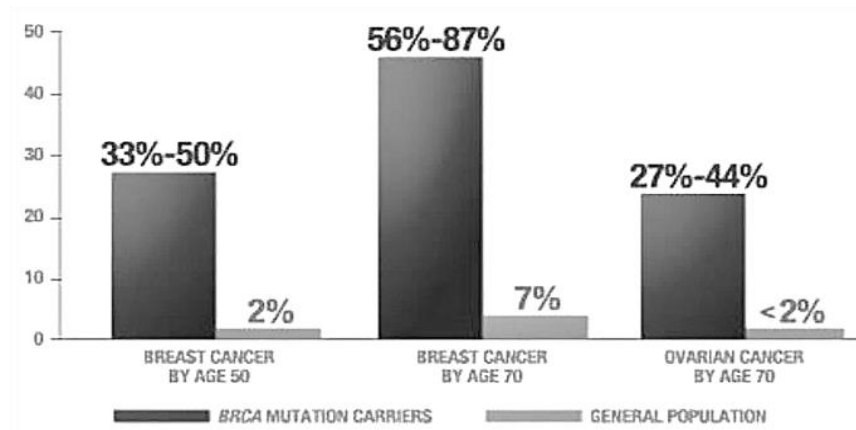
|          | <b>Stem cells</b>                                            | <b>Cancer cells</b>                                                                           |
|----------|--------------------------------------------------------------|-----------------------------------------------------------------------------------------------|
| <b>1</b> | Differentiate into specialised cell types;                   | Do not differentiate into specialised cell types / remain undifferentiated and unspecialised; |
| <b>2</b> | Experience contact inhibition;                               | Does not experience contact inhibition;                                                       |
| <b>3</b> | Experience anchorage dependence / does not metastasize;      | Does not experience anchorage dependence / metastasize to set up secondary tumors;            |
| <b>4</b> | Divides or stops dividing upon response to molecular signals | Divides indefinitely / does not respond to molecular signals;                                 |
| <b>5</b> | AVP: i.e. normal proto-oncogene / tumor suppressor gene      | AVP: i.e. mutated proto-oncogene / tumor suppressor gene                                      |

Hereditary breast and ovarian cancer (HBOC) syndrome is an inherited condition that causes an increased risk of developing these cancers, often before the age of 50.

Research shows that about 10% of breast and ovarian cancer is due to an alteration in either of 2 genes: BRCA1 and BRCA2. These genes can come from either your mother or your father.

BRCA1 is a human tumour suppressor gene (also known as a caretaker gene) and is responsible for repairing DNA. BRCA1 and BRCA2 are unrelated proteins, but both are normally expressed in the cells of breast and other tissue, where they help repair damaged DNA, or destroy cells if DNA cannot be repaired.

The figure below shows data concerning breast cancer as a result of mutations in the BRCA genes 1 & 2 on chromosome 17 and 13 respectively. Ovarian cancer data shows a link to BRCA cancer especially by age 70.



- (b) Explain the difference in percentage occurrence of breast cancer between woman who are BRCA mutation carriers and woman in the general population by the age of 50.

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..... [3]

1. Women who are *BRCA* mutation carriers had a **higher percentage** of occurrence of breast cancer by **31% - 48% higher** as compared to women in the general population.
2. *BRCA* genes are **tumor suppressor genes**. *BRCA* mutation carriers having inherited mutated *BRCA* genes will be at an **increased risk of developing breast cancer**.
3. **Loss of function of both alleles** of *BRCA* genes leads to **non-functional BRCA proteins**.
4. **BRCA proteins are involved in DNA repair**, and with the loss of function there is an **increase in cells with DNA mutated**.
5. **Accumulation of mutations** in these cells may result in a Gain of function mutations e.g. proto-oncogenes to oncogenes OWTTE.

[Total: 10]

- 5 In a dihybrid inheritance, gene B/b codes for flower colour while gene H/h codes for leaf shape of a plant.

The F1 progeny of a pure-bred plant with red flowers and oval leaves, and another pure-bred plant with yellow flowers and fan-shaped leaves, have red flowers and fan-shaped leaves.

F1 plants then **undergo a test cross**.

- (a) Predict the expected phenotypic ratio in the offspring progeny.

.....[1]

1 Red flower, fan-shaped leaf : 1 Red flower, oval leaf : 1 Yellow flower fan-shaped leaf : 1 yellow flower, oval leaf

*[Reject: 1:1:1:1 with no phenotypes]*

- (b) Identify and explain the dominant traits in this dihybrid cross.

.....  
 .....  
 .....  
 ..... [2]

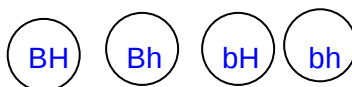
- 1 red flowers and fan-shaped leaves;
- 2 ref. **dominant traits / alleles** being expressed in the **phenotype** in the **heterozygous** condition (i.e. F1 progeny) / **recessive alleles** (coding for yellow flowers and fan-shaped leaves), **being masked by dominant alleles** in the **heterozygous** condition;

- (c) Using the symbols for the alleles stated above, draw a genetic diagram to show the expected phenotypic ratios for the offspring of the test cross.

F<sub>1</sub> phenotypes:                      Red flower,                      x                      Yellow flower,  
                                          Fan-shaped leaf                                              Oval leaf

F<sub>1</sub> genotype:                      BbHh                                              bbhh

F<sub>1</sub> gametes



F<sub>2</sub> genotypes:

Punnett square:

|                                                                                   |  |  |  |  |
|-----------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|
|  | BbHh<br>(Red flower, Fan-shaped leaf)                                             | Bbhh<br>(Red flower, Oval leaf)                                                   | bbHh<br>(Yellow flower, Fan-shaped leaf)                                            | bbhh<br>(Yellow flower, oval leaf)                                                  |

F<sub>2</sub> / Progeny phenotypes:    Red flower, fan-shaped leaf    Red flower, Oval leaf    Yellow flower, Fan-shaped leaf    Yellow flower, oval leaf

F<sub>2</sub> / Progeny phenotypic ratio:                      1                      1                      1                      1

[L2] Source: Novel

**Mark scheme:**

1. F<sub>1</sub> phenotype and genotypes
2. Parental gametes – (Gametes **must** be circled)
3. F<sub>2</sub> genotypes correspond to phenotypes

[3]



- (d) Plants are a good choice of experimental organisms for carrying out such crosses and for performing statistical tests.

Compared to plants, humans are less ideal and it is usually more difficult to arrive at reliable conclusions for observations involving humans. Suggest why.

.....

.....

..... [1]

1. Humans produce **limited offspring**. (This makes statistical tests difficult)
2. Humans have a long life span and **some traits only appear at a later stage in life**

*[Reject: some human traits are under continuous variation. Plants also have continuous variation]*

[Total: 7]

**Section B**  
**Answer EITHER 6 OR 7.**

Write your answers on the separate answer paper provided.  
Your answers should be illustrated by large, clearly labelled diagrams, where appropriate.  
Your answers must be in continuous prose, where appropriate.  
Your answers must be set out in sections **(a)**, **(b)** etc., as indicated in the question.

- 6 (a)** The expression of genes gives rise to proteins that affect the biochemical reactions and other processes in organisms.

Discuss, with examples, the importance of specific shapes of proteins in organisms. [8]

1. Active site of enzyme has specific shape that substrate can fit into
2. Via lock and key mechanism
3. *[Importance]* To form enzyme-substrate complex/ products important for metabolic pathways / increase the rate of reaction
4. *[Example]* any enzyme and substrate
  
5. Binding site of protein complementary to conformation of DNA region to which the protein binds (point 6-9, 16-19)
6. *[Importance]* Ref. to **DNA replication**
7. *[Example]* DNA polymerase binds to 3'OH group of existing nucleotide/ helicase binds to origin of replication/ topoisomerase binds to DNA to unzip the double-stranded
8. *[Importance]* Ref. to **transcription**
9. *[Example]* **transcription factor** binding to DNA / **RNA polymerase** binds to promoter
  
10. Binding of substances to transport proteins
11. *[Importance]* Allows for movement of substances across cell membrane
12. *[Example]* transmembrane protein e.g. Na<sup>+</sup> channel, Na<sup>+</sup>K<sup>+</sup> pump, glucose transporter etc
  
13. *[Example]* Haemoglobin is made up of **4 polypeptides** and their **haem groups** to form a specific conformation
14. *[Importance]* allows it to **bind to oxygen molecules** to form oxyhaemoglobin/ transport oxygen to all parts of the body
15. Reference to **cooperative binding**
  
16. *[Importance]* Ref. to **separation of sister chromatids during anaphase**
17. E.g. kinetochore to bind to centromere via complementary shape
  
18. *[Importance]* Ref. to **amino acid activation**
19. E.g. tRNA anticodon bind to the anticodon attachment site of amino-acyl tRNA synthetase

QWC [1]

Importance of specific shapes communicated accurately and to include at least two examples

(b) Compare and contrast DNA replication with transcription.

[7]

Similarities between transcription and replication:

Both occur in the nucleus;

Both processes involves the synthesis of polymers from monomers;

Both need template to initiate the synthesis of product;

Both processes involves complementary base pairing;

Both processes require energy from ATP;

[Max: 3]

| Feature                                                             | DNA replication                                                                                                                                                                 | DNA transcription                                                                                                  |
|---------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------|
| Nucleotide bases involved                                           | Deoxyribonucleotides; newly synthesised strand complementary to the template strand where A pairs with T and C pairs with G;                                                    | Ribonucleotides; newly synthesised mRNA strand, complementary to template strand, where A pairs with U and C to G; |
| End-product                                                         | 2 double stranded DNA molecules, formed by semi-conservative method: two molecules of DNA formed: each consist of one newly synthesized daughter strand and one parental strand | 1 single stranded mRNA molecule is complementary to the template DNA strand<br><br>[Accept: tRNA or rRNA]          |
| Number of strands required for synthesis of complementary strand(s) | Both DNA strands of double helix DNA molecule are required to act as template strand for synthesis of complementary strands                                                     | Only one strand of DNA molecule is required to act as template strand for the synthesis of mRNA                    |
| Where enzymes bind for start of process                             | Origin of replication where helicase binds to: unwind and unzip the double stranded DNA                                                                                         | Promoter region, near 3' end of gene where RNA polymerase binds to: unwind and unzip the DNA                       |
| Enzymes used in producing the complementary strand                  | DNA polymerase III is used in synthesising the complementary strand by adding DNA nucleotides to the 3' end of the existing strand (primer)                                     | RNA polymerase II is used in transcribing the pre-mRNA strand, by adding nucleotides to 3' end of strand           |
| Primers requirement                                                 | Need primers for the 3' end (from existing strand) as DNA Pol is unable to initiate polynucleotide synthesis                                                                    | Do not require primers as RNA pol is able to initiate polynucleotide synthesis                                     |

[Max: 4]

[Total: 15]

- 7 (a) Biochemical reactions in cells often involve cyclic metabolic pathways which occur within compartments inside cells.

Compare the Krebs cycle and Calvin cycle in plant cells **and** suggest why the **cellular compartmentalisation** in both cycles is so important. [7]

|                                   | Krebs cycle                                                                              | Calvin cycle                                                    |
|-----------------------------------|------------------------------------------------------------------------------------------|-----------------------------------------------------------------|
| 1. CO <sub>2</sub>                | Produces CO <sub>2</sub>                                                                 | Uses CO <sub>2</sub>                                            |
| 2. ATP                            | Produces (small quantity) of ATP                                                         | Uses ATP                                                        |
| 3. Molecule regenerated           | Oxaloacetate                                                                             | RuBP / ribulose biphosphate                                     |
| 4. Molecule that enters the cycle | Acetyl-coA                                                                               | CO <sub>2</sub>                                                 |
| 5. Coenzymes / reducing powers    | Produces reduced NAD/NADH & reduced FAD/FADH <sub>2</sub>                                | Uses reduced NADP / NADPH                                       |
| 6. Nature                         | Catabolic / oxidising                                                                    | Anabolic / reducing                                             |
| 7. Energy conversion              | Chemical energy (carbon intermediates) to chemical energy (ATP, NADH/FADH <sub>2</sub> ) | Chemical energy (ATP and NADPH) to chemical energy (GP/glucose) |
| 8. Enzymes                        | Both involves enzymes                                                                    |                                                                 |

[Max: 4]

At least one of the following:

9. **Idea that** isolation of enzymes and reactants from the rest of the cell...
10. **Idea that** ...so to allow reactions to occur in small volume / high concentrations
11. **Idea that** ensure maintenance of optimal conditions for specialised biochemical reactions to occur
12. **Idea that** mitochondria can be streamed to areas needing ATP / movement of organelles within cytoplasm.
13. (**Idea that** Keeps products separate from cytoplasm by use of membranes.)

[Max 4]

- (b) Discuss briefly the role of meiosis in evolution **and** explain, with examples, how environmental factors act as forces of natural selection.

[8]

Description of how meiosis leads to genetic variation:

1. Crossing over between homologous chromosomes during prophase I;
2. Independent assortment of chromosomes during metaphase and anaphase (I and II);
3. producing genetically varied gametes;
4. Leading to formation of genetically varied organisms due to random fusion of gametes;

Max 3m

Impact of meiosis on evolution:

5. Meiosis produces new combination of alleles;
6. Lack of genetic variation may lead to death of all members in the population due to the selection pressure present;
7. For natural selection to occur, there must be a varied gene pool / presence of different alleles;
8. In the presence of an environmental change, the environment will select the variants with selective advantage
9. These variants will be able to survive, reproduce and pass on their alleles to their offspring in greater numbers;
10. Leading to a change of allele frequency in population (evolution).
11. Mutations are the key source of genetic variation by producing new alleles and not merely new combination of alleles;
12. Mutation therefore has a more significant role in evolution compared to meiosis;

Max 4m

Examples of how environmental factors act as forces of natural selection.

For each example:

- 1 a **named example** of a species that has evolved in this way;
- 2 description/clear statement of the **change** that occurred in the environment / **Selection pressure**;
- 3 description/clear statement of **different varieties** (that existed at the same time);
- 4 **explanation** of/reason for one variant having a **selective advantage**;
- 5 the **change in the population/species** due to **natural selection/evolution**;

**Example 1: Galapagos Finches (Darwin's finches)**

- 1 For natural selection to occur, there must be heritable variation for a particular trait. In this case, it is the alleles of the gene which determine the size and depth of the beaks in Galapagos finches.
- 2 Give **e.g. of variation** - Some had large and heavy beaks adapted for eating large seeds, others for small seeds; some had parrot-like beaks for feeding on buds and fruits, and some had slender beaks for feeding on small insects. One used a thorn to probe for insect larvae in wood, like some woodpeckers do. (Six were ground-dwellers, and eight were tree finches.)
- 3 **Selection pressure: Limited food source**

- 4 The type of beak phenotype that is being **selected for** depends on the availability of the food source due to different environmental conditions or different habitats.
- 5 E.g. If small tender seeds are available at that island, finches with small beaks are at a **selective advantage** as it allows it to feed. If large hard seeds are available at that island, finches with large, more powerful beaks were **selected for / selective advantage**.
- 6 Individuals which are more adapted to surviving in a particular habitat will **survive to maturity, reproduce to produce viable offspring** and **pass on the beneficial alleles** to the next generation.
- 7 Hence there was **differential survival and reproductive success** associated with the possession of the particular beak type, therefore this leads to a change in **allele frequency** in a population for beak type.

### Example 2: Soapberry bugs (*Jadera haematoloma*)

- 1 **Heritable variation** - beak length. Soapberry bugs feed most effectively when their beak length closely matches the depth at which the seeds are found within the fruit.
- 2 **Selection pressure: Change in food supply**
- 3 Food supply - the soapberry bug feeds on the seeds of a native plant, the balloon vine (*Cardiospermum corindum*). However in Central Florida, balloon vines have become rare and thus the soapberry bugs in this region feeds on the goldenrain tree (*Koeleruteria elegans*), a species that was introduced from Asia.
- 4 Seeds of the goldenrain tree fruit are much closer to the surface than seeds of the balloon vine. Therefore bugs with shorter beak lengths would be selected for by natural selection, as they would be able to feed on the seeds of goldenrain tree fruit, which are more widely available.
- 5 In Southern Florida where the balloon vine is more common, the seeds are found deeper within the fruit. Therefore bugs with longer beak lengths would be selected for by natural selection, as they would be able to feed on the seeds of balloon vine fruit, which are more widely available.
- 6 Individuals which are more adapted to feeding on the seeds of the plant at the specific region will survive to maturity, reproduce to produce viable offspring and pass on the beneficial alleles to the next generation.
- 7 Hence there was differential survival and reproductive success associated with the possession of the particular beak length, therefore this leads to a change in allele frequency in a population for beak length at that region. For central Florida, allele frequency for shorter beak length increased due to natural selection, over successive generations. For southern Florida, allele frequency for longer beak length increased due to natural selection, over successive generations.

### Example 3: Evolution of drug-resistant bacteria

**(MRSA: methicillin-resistant *Staphylococcus aureus*)**

*Staphylococcus aureus*/MRSA/*Clostridium difficile*/other named species;

Selection pressure: Use of methicillin antibiotic ;

some bacteria were resistant and others were not;

resistant bacteria survived (and multiplied) while non-resistant were killed;

percentage of the population showing resistance increased;

#### QWC:

At least 2 examples of natural selection including the respective type of environment factor acting as the force of natural selection.

**END OF PAPER**